SHRIHARI M V

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Professional Summary

Applied Machine Learning and Data Science undergraduate with hands-on experience building end-to-end AI systems across **Computer Vision**, **Predictive Analytics**, and **Retrieval-Augmented Generation**. Experienced in dataset curation, feature engineering, model evaluation, experiment tracking, explainability, calibration, deployment, and production inference using **PyTorch, Scikit-learn, FastAPI and MLflow**. Passionate about developing reliable, interpretable, and scalable AI solutions.

Technical Skills

Programming	Python, SQL, Java, C
Machine Learning	Scikit-learn, XGBoost, Feature Engineering, Cross Validation, SMOTE, Model Evaluation, Calibration
Deep Learning	PyTorch, TensorFlow, Swin Transformer, Transfer Learning, OpenCV, Albumentations, Focal Loss, Explainable AI
Generative AI	Transformers, Sentence Transformers, CrossEncoder, RAG, Semantic Search, Vector Databases
Data Analysis	Pandas, NumPy, EDA, Statistical Analysis, Power BI, Matplotlib
Deployment	FastAPI, Docker, MLflow, Weights & Biases, Gradio, Streamlit
Tools	Git, GitHub, VS Code, Google Colab, ChromaDB, Ollama

Projects

AksharaVision – Kannada Handwritten Character Recognition System GitHub

Python • PyTorch • Swin Transformer • OpenCV • FastAPI • Gradio • TinyDB • ONNX Runtime

- Curated a balanced **56.5K-image, 113-class** Kannada OCR dataset using OpenCV preprocessing, augmentation, and quality-validation pipelines for robust model training.
- Benchmarked CNN, EfficientNet, and Swin Transformer architectures; selected **Swin V3** with a **512-dimensional embedding head** through iterative experimentation.
- Improved prediction reliability through temperature-scaled confidence calibration, achieving **Macro-F1 0.9989**, **ECE 0.00065**, and **Brier Score 0.00112**, supported by systematic error analysis.
- Investigated failure modes using confusion-pair analysis, hard-sample diagnostics, robustness evaluation, occlusion-sensitivity explainability, and **Weights & Biases** experiment tracking.
- Developed FastAPI and Gradio inference workflows with Top-3 predictions, telemetry logging, human feedback collection, PSI-based drift monitoring, and **ONNX Runtime** export for deployment portability.

AttriShield – Employee Attrition Prediction System GitHub

Python • Scikit-learn • XGBoost • FastAPI • MLflow • SHAP • SQL • Streamlit

- Built an employee attrition decision-support system using HR records from **1,470 employees** and **35+ features** to prioritize retention interventions.
- Engineered a leakage-safe feature pipeline with redundancy analysis, **SMOTE**, cost-aware threshold tuning (**0.15**), and stratified **5-fold cross-validation** for robust model evaluation.
- Benchmarked Logistic Regression, Decision Tree, Random Forest, and **XGBoost** using **PR-AUC**, **ROC-AUC**, Recall, and F1; selected XGBoost achieving **0.625 CV PR-AUC**.
- Validated model reliability using **SHAP**, calibration (**Brier Score 0.105**, **ECE 0.056**), and **MLflow** tracking; delivered FastAPI batch inference and Streamlit dashboards to prioritize high-risk employees and support HR retention planning.

CarnaticPrep – Retrieval-Augmented Knowledge System GitHub

Python • ChromaDB • Sentence Transformers • CrossEncoder • Ollama • FastAPI • RAGAS

- Designed a modular hybrid RAG pipeline combining dense retrieval, BM25, adaptive query routing, weighted Reciprocal Rank Fusion (RRF), and CrossEncoder reranking for domain-specific knowledge retrieval.
- Built corpus-aware indexing pipelines using structure-aware chunking, schema-aware metadata normalization, and adaptive PDF/CSV balancing to improve retrieval across heterogeneous knowledge sources.
- Evaluating retrieval quality using **RAGAS** on a benchmark query set to optimize retrieval effectiveness.
- Developed an end-to-end FastAPI-powered RAG system supporting vector indexing, configurable retrieval workflows, structured logging, Swagger/OpenAPI documentation, and grounded response generation using Ollama.

Education

B.E. Computer Science & Engineering (Data Science) 2022 – 2026
Bangalore Institute of Technology CGPA: **8.6/10**

PCMB 2020 – 2022
Sri Kumarans PU College Percentage: **94.83%**

Experience

MindMatrixEd – Generative AI Intern Feb 2026 – May 2026

- Explored AI-assisted Android application workflows using Google AI Studio, Android Studio and Kotlin contributing to feature prototyping and user-centric product design.